

Numerical optical optimization of monolithic planar perovskite-silicon tandem solar cells with regular and inverted device architectures

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Abstract: We numerically maximize the achievable photocurrent density of planar perovskite-silicon tandem solar cells for different device architectures. For the optimizations we combine the transfer-matrix method with a simulated annealing algorithm. The optimizations are conducted within experimentally accessible and relevant layer-thickness ranges, which allows to extract applicable device guidelines. A comparison between regular and inverted tandem-cell designs reveals that a rear-emitter silicon heterojunction in combination with an inverted perovskite top-cell can yield a photocurrent, which is 1.4 mA/cm² higher than that of tandem cells with the usual polarity and a front-emitter silicon bottom cell. Switching from the regular to the inverse architecture leads to over 2% (absolute) gain in power conversion efficiency. Finally we show that an efficiency of 30.8% is achievable for such tandem cells with an optimized perovskite band-gap.

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1. Introduction

Probably the most important technological limit for solar cells is the Shockley-Queisser (SQ) limit, which states that the power conversion efficiency (PCE) of single-junction solar cells is limited to slightly above 30% [1]. During the last decades many so-called *third-generation* concepts were investigated in order to overcome the SQ limit, but the losses in most concepts hindered them from exploiting their full potential. So far only multi-junction solar cells could surpass the SQ limit [2].

In multi-junction solar cells, two or more solar cells with different bandgaps are stacked onto each other in a way that the *top cell*, which the incident sunlight hits first, has the highest bandgap, and every subsequent cell has a lower or similar bandgap than the previous one. Hence, photons with energies E_{ph} above the bandgap energy E_g of a subcell are (likely) absorbed in that cell while lower energy photons may propagate into the cells with lower bandgaps. This design helps to reduce thermalization losses, which arise especially for high energy photons because only a maximum energy fraction E_g/E_{ph} can be utilized by a solar cell with absorber bandgap E_g .

Because of their excellent material quality, low cost and high market penetration, wafer-based crystalline silicon (c-Si) solar cells are a canonical choice for the low-bandgap bottom cell of a multi-junction solar cell. For back-contacted silicon-heterojunction (SHJ) solar cells with intrinsic and doped hydrogenated amorphous silicon layers on both sides, a record PCE of 26.33% was achieved [2]. Hence, 89% of the theoretical PCE limit for single-junction silicon solar cells (29.4%; see [3]) were achieved and further improvement of this technology will become more and more difficult.

Several theoretical studies revealed that lead halide perovskites are the most promising candidates for the top-cell material of a double-junction (*tandem*) solar cell with a c-Si bottom cell to date [4,5]. In recent years, perovskite solar cells have shown an astonishing rise in performance with a current certified PCE record of 22.1% [2], quickly leading to the development of highly efficient monolithic perovskite/silicon tandem cells with efficiencies up to 23.6% [6–8]. In addition, numerous detailed numerical studies of the potential of perovskite-Si tandem devices were presented [9–13].

In 2015, Albrecht and co-workers presented the first monolithic tandem solar cell consisting of a perovskite top cell and a silicon heterojunction (SHJ) bottom cell with 19.9% PCE [6]. One issue that hindered this cell from surpassing the PCE of high-end single-SHJ solar cells was the *current mismatch* between the top and the bottom cells: highly efficient tandem solar cells require well-matched photocurrent densities arising from the absorption spectra of the top and bottom solar cells, because the overall current density is controlled by the lowest of the two.

Werner and co-workers improved current matching by reducing the perovskite absorber thickness, such that more light is transmitted into the limiting silicon bottom cell, and by implementing a textured anti reflection foil [7]. However, a significant portion of the incident light was still parasitically lost in the contact layers, e.g. in the hole-transport contact of the perovskite top cell.

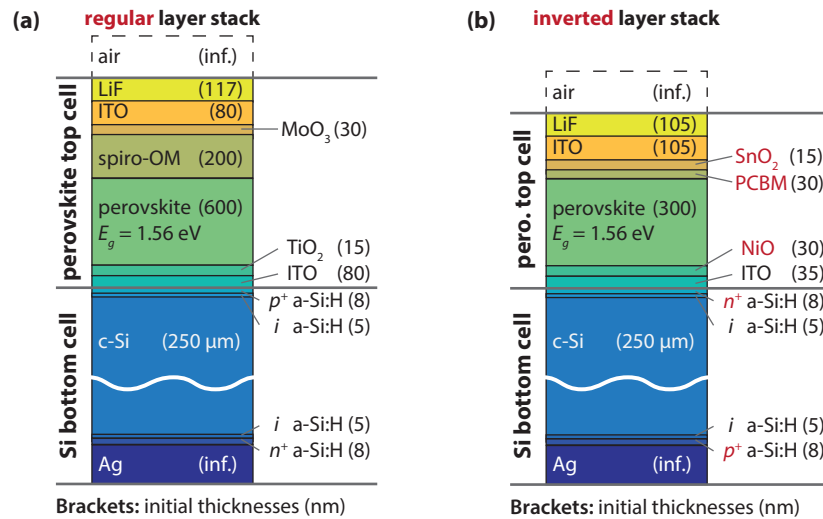


Fig. 1. The layer stacks used for the optical simulations: (a) the regular architecture with the hole-selective and p -layers on the front of the perovskite and silicon subcells, respectively, and (b) the inverted architecture with the electron-selective and n -layers on the front of the perovskite and silicon subcells, respectively. The numbers in brackets refer to the layer thicknesses used as starting values in the optimizations.

In a recently published study we presented an optimization of the top-cell layer thicknesses of a perovskite-SHJ tandem solar cell from an optical point of view [14]. From that study we learned that parasitic absorption losses of the front hole-transporting layer (Spiro-OMeTAD) always exceed 2.5 mA/cm^2 for realistic film properties and thicknesses. In addition, the usually applied MoO₃ buffer layer within the hole top contact suffers from degradation under sputter-deposition of the top TCO, which increases the parasitic absorption even further, as shown by Werner and co-workers [7].

Lately Bush and co-workers presented a monolithic tandem cell at 23.6% efficiency with an inverted polarity: a rear emitting silicon heterojunction was combined with an inverted perovskite top cell that is characterized by an electron-selective top contact [8]. Here the parasitic absorption was strongly reduced, because they were able to make the supporting layers of the perovskite subcell very thin, leading to high photocurrent densities in both subcells. In addition, when implementing very thin metal oxide buffer layers such as SnO₂ formed by atomic layer deposition (ALD) [15], the parasitic absorption can be further reduced with respect to regular hole-selective top contacts used by Albrecht and Werner, which are based on Spiro-OMeTAD contacts.

Thus, to further increase the tandem cell efficiency optical optimizations based on experimentally relevant material stacks and thickness combinations are highly necessary. In this study, we therefore extend the optical optimizations presented earlier [14] to an *inverted architecture*, where both the top perovskite and bottom silicon cells are flipped upside down in polarity as compared to “regular” structures. As Grant and co-workers pointed out, *a priori* neither of the two architectures is superior from an optical point of view [13]. However, the inverted architecture has the great advantage of electron-selective contacts that can be processed much thinner resulting in less parasitic absorption and hence higher photocurrent densities. To allow an easy comparison, we also briefly summarize the original results as they were presented at the 2016 OSA Light, Energy and the Environment Congress [16].

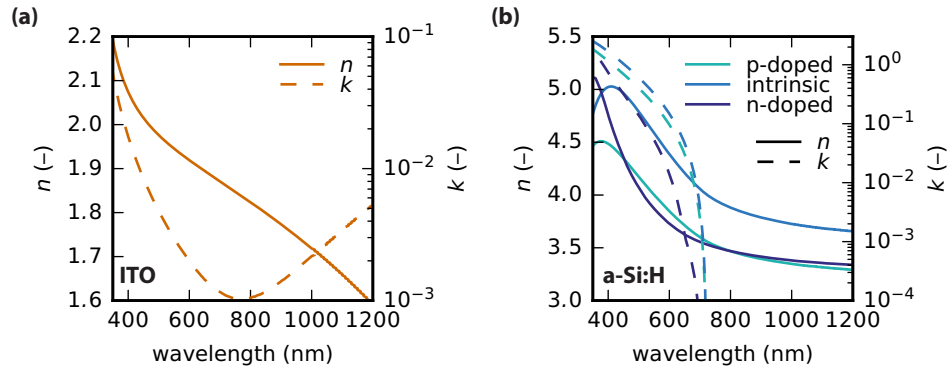


Fig. 2. The (n, k) data for (a) the indium tin oxide (ITO) layer and (b) the different amorphous hydrogenated silicon (a-Si:H) layers.

2. Simulated optical systems

Figure 1 illustrates the two device architectures with experimentally relevant layer stacks considered for this study. The regular layer stack is shown in Fig. 1(a): its top cell consists of a lithium fluoride (LiF) antireflective coating, an indium tin oxide (ITO) front contact, a molybdenum oxide (MoO_3) buffer layer, a hole-selective contact layer (Spiro-OMeTAD), the perovskite absorber, an electron-selective titanium oxide (TiO_2) layer, and an ITO interconnecting layer. The silicon heterojunction (SHJ) bottom cell consists of p^+ -doped and intrinsic (i) hydrogenated amorphous silicon (a-Si:H) layers, the c-Si wafer, i and n^+ -doped a-Si:H layers and the silver (Ag) back reflector and contact.

Figure 1(b) sketches the *inverted* layer stack: here, the top cell consists of a lithium fluoride (LiF) antireflective coating, an indium tin oxide (ITO) front contact, a tin oxide (SnO_2) buffer layer, an electron-selective contact layer (PCBM), the perovskite absorber, a hole-selective nickel oxide (NiO) layer, and an ITO interconnecting layer. The silicon heterojunction (SHJ) bottom cell consists of n^+ -doped and intrinsic (i) hydrogenated amorphous silicon (a-Si:H) layers, the c-Si wafer, i and p^+ -doped a-Si:H layers and the silver (Ag) back reflector and contact.

The complex refractive index spectra (n, k) for the ITO layers [17], the intrinsic a-Si:H layers [18] and the doped a-Si:H layers [19] were derived using reflection-transmission measurements and are shown in Fig. 2. The (n, k) data of the other layers were taken from literature: LiF [20], spiro-OMeTAD [9], perovskite [21], MoO_3 [22, 23], NiO [24], spray-deposited TiO_2 [25], and SnO_2 [26]. Note that the SnO_2 data were taken for fluorine-doped APCVD material, while experimentally an intrinsic ALD SnO_2 layer would be used as buffer layer (see [15], supplementary). Hence, the parasitic absorption in the SnO_2 layer is slightly overestimated in the simulations. However, this safely can be ignored because the layers are very thin.

3. Method

The **absorption profiles** were calculated with the MATLAB-based program GenPro4, which is developed at the Delft University of Technology [27, 28] and utilizes the **transfer-matrix method** (see for example [29] 1.6 and 7.6.1.). The layer stack constituting the simulation domain is enclosed by incoherent infinitely-thick air and silver layers, respectively. Further, the c-Si wafer was treated incoherently because its thickness of 250 μm is exceeding the coherence length of sunlight by far. All the other layers were treated coherently and **all interfaces were considered to be optically flat, i.e. they do not scatter any light**. From the absorption profiles of the different

layers current densities were calculated with

$$J_{\text{ph}, i} = -e \int_{350 \text{ nm}}^{1200 \text{ nm}} A_i(\lambda) \Phi(\lambda) d\lambda, \quad (1)$$

where e is the elementary charge, A_i is the absorption spectrum of the i -th layer, Φ is the photon flux according to the AM1.5G solar spectrum [30] and λ denotes the wavelength. The maximum achievable current densities for the top and bottom cells are calculated by using the spectra A_i of the perovskite and c-Si layers, respectively. The current densities for all the other layers are losses due to parasitic absorption.

For both architectures, we conducted layer thickness optimizations for the 7 layers constituting the perovskite top cell (see Fig. 1), using the simulated annealing algorithm [31], as it is implemented in the MATLAB Global Optimization Toolbox. For the optimization we used the realistic start, minimum and maximum thicknesses shown in Table 1. The very thin thicknesses used for the inverse architecture are justified by recent results from Bush and co-workers [8]. The goal of the optimization was to maximize the current matching point, i.e. the current density of the limiting subcell,

$$\min(J_{\text{ph}}^{\text{top}}, J_{\text{ph}}^{\text{bottom}}) \rightarrow \max. \quad (2)$$

During the optimization, the thicknesses of the layers constituting the SHJ bottom cell were kept constant.

Table 1. The experimentally relevant initial, minimum and maximum thicknesses of the top perovskite cell as they were used in the optical simulations, and the set of optimal thicknesses used for Fig. 3. The layer thicknesses of the bottom silicon heterojunction solar cell were kept constant and are depicted in Fig. 1. All values are in nm.

<i>regular</i>	start	min.	max.	opt.	<i>inverse</i>	start	min.	max.	opt.
LiF	117	0	200	107	LiF	105	80	130	104
ITO	80	60	150	103	ITO	105	80	130	80
MoO ₃	30	20	40	27	SnO ₂	15	5	30	5
Spiro-OM	200	150	300	150	PCBM	30	10	60	10
<i>perovskite</i>	600	200	800	300	<i>perovskite</i>	300	200	400	286
TiO ₂	15	10	20	11	NiO	30	10	50	28
ITO	80	20	100	20	ITO	35	20	50	20

4. Results

Figure 3(a) shows one of the obtained optima for the regular architecture with the thicknesses from Table 1. Because of its pseudorandom nature, the simulated annealing algorithm does not find *the* global optimum, but solutions in its close proximity. All the runs led to optima, which were very close to each other. We see that the optimum of $J_{\text{ph}} = 17.6 \text{ mA/cm}^2$ is at the current-match condition, because

$$\min(J_{\text{ph}}^{\text{top}}, J_{\text{ph}}^{\text{bottom}})$$

is maximal when the available current is distributed equally between the top cell and the bottom cell. The **largest loss is the reflective loss** with 7.69 mA/cm^2 . The largest parasitic absorption occurs in the hole-transporting layer (Spiro-OMeTAD) and is equivalent to 2.61 mA/cm^2 . In the front ITO and MoO₃ layers 0.36 mA/cm^2 and 0.15 mA/cm^2 are lost, respectively. In all the other layers the losses are 0.06 mA/cm^2 or lower. Note that the reflectivity has a pronounced peak around 870 nm. **This peak arises from interference in the thin-film stack covering the Si wafer.** Depending on the deposition conditions, ITO can be much more absorptive as

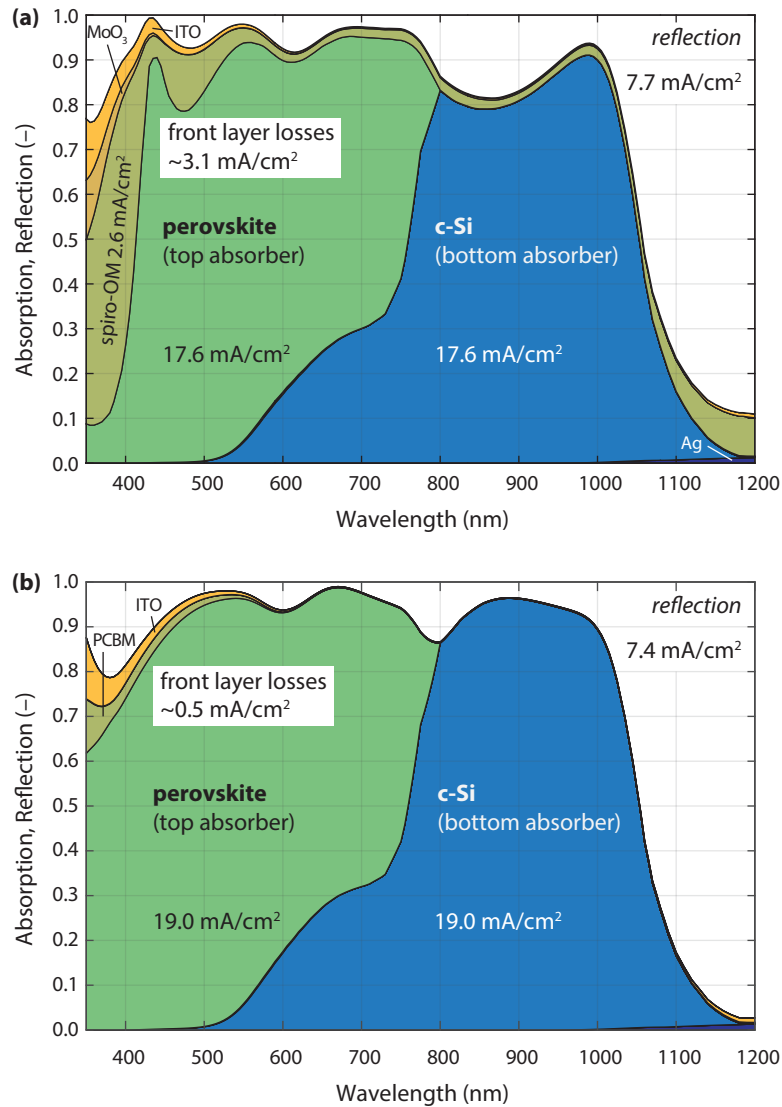


Fig. 3. Absorption profiles of optimized devices for (a) the regular architecture, and (b) the inverted architecture. Mainly because of the high parasitic absorption in Spiro-OMeTAD, with the inverse architecture almost 2.8 mA/cm² can be gained with respect to the regular architecture.

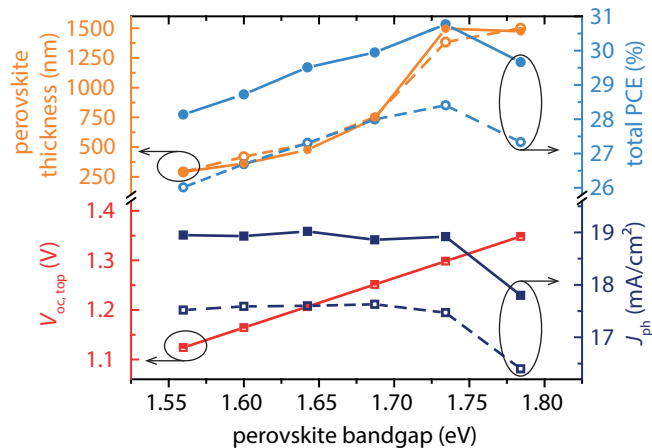


Fig. 4. Estimating the tandem cell efficiency of regular (dashed lines) and inverted (solid lines) device architectures as a function of the perovskite bandgap. The following parameters are shown: the open-circuit voltage (V_{oc}), being similar for both architectures, the optimized absorber thickness of the perovskite top-cell as well as the maximum achievable current density J_{ph} , and the tandem-cell power conversion efficiency (PCE). For the PCE calculation a FF of 81% and a silicon V_{oc} of 709 mV in tandem geometry was assumed. The perovskite (n , k) data for the different bandgaps was obtained by blue-shifting the original data.

compared to the data utilized here [13]. However, potentially ITO can be replaced by novel TCO materials such as hydrogen-doped indium oxide ($\text{In}_2\text{O}_3\text{:H}$) [32] or indium zinc oxide (IZO) [33].

Figure 3(b) shows an optimum for the inverse architecture; the corresponding thicknesses are shown in Table 1. Here, the optimum is at $J_{ph} = 19.0 \text{ mA/cm}^2$. Hence, with respect to the regular architecture about 2.8 mA/cm^2 can be additionally collected in the absorber layers. This gain is mainly caused by omitting the necessarily thick Spiro-OMeTAD layer, which reduces the total parasitic absorption losses to about 0.6 mA/cm^2 . Further, because of the much thinner layers covering the Si wafer, the broad reflectance peak around 870 nm is not present in the inverse structure and a larger fraction of light can be coupled into the wafer.

The optimizations led to thicknesses of the top contact layers, i.e. ITO, SnO_2 and PCBM, which are very close to the experimentally relevant lower boundaries (see Table 1). Thus, in order to maximize the photocurrent density, these layers need to be processed as thin as possible, while still ensuring good current collection for ITO, sufficient buffer protection against sputtering for the SnO_2 and a good electron contact formed by PCBM layer.

To see whether the superiority of the inverse architecture is solely caused by its very thin supporting layers or whether also the optical constants of the different layers support a better coupling of the incident light into the solar cells, we performed another simulation: we combined the regular layer stack depicted in Fig. 1(a) with the layer thicknesses of the optimized inverse layer stack as in Table 1. Only the perovskite thickness was increased from 286 nm to 290 nm for current matching. This leads to a current density of 19.0 mA/cm^2 for both the perovskite and the silicon layers. Hence, if it was possible to process a regular solar cell with as thin supporting layers as for the inverse architecture, its optical performance would be comparable, as suggested by Grant and co-workers [13]. Further, the highly absorbing Spiro-OMeTAD could potentially be replaced by PTAA as a hole-transporting material [34, 35].

In [14] we studied, how shifting the perovskite bandgap E_g affects the overall device performance for the regular structure. For the present study we repeated this procedure for the inverse architecture. To estimate the PCE of the simulated devices, we reduced the open circuit voltages

(V_{oc}) of state-of-the-art single-junction perovskite and SHJ cells in order to take into account that V_{oc} is logarithmically dependent on J_{ph} . Hence, the open circuit voltage of the top cell with the “standard” perovskite ($E_g = 1.56$ eV) was set to $V_{oc, top} = 1.13$ V and for the Si bottom cell we arrived at $V_{oc, bottom} = 0.709$ V. These values include reductions of 4 mV (6 mV) and 19 mV (21 mV) for the top and bottom cells in the inverted (regular) architecture, respectively. The optical parameters of the higher band-gap perovskites were generated by shifting the optical constants from [21] into the blue in steps of 20 nm. Blue-shifting the complete perovskite (n, k) data is justified by experimental findings [36]. We assumed $V_{oc, top}$ to increase linearly with the bandgap. The fill factor of the tandem device was estimated to be 81% [14].

Figure 4 shows the results of the bandgap optimization for the inverted structure (solid lines) and for the regular structure (dashed lines). Increasing the perovskite bandgap does not affect the optimized current density, except for the largest bandgap (1.78 eV), where the perovskite thickness required for current matching exceeds the experimentally reasonable maximum of 1500 nm [37, 38]. At such large perovskite bandgaps current matching can be hardly achieved even if thicker perovskite layers would be allowed. At the optimal perovskite bandgap of 1.73 eV, $V_{oc, top} \approx 1.3$ V, leading to a total open-circuit voltage of 2.01 V, and $J_{ph} = 18.92$ mA/cm². These values indicate a feasible maximum power-conversion efficiency of 30.8%. The optimized efficiency occurs at similar bandgap values for the inverted and regular architecture [14] and is in accordance with [39], where the optimal top-cell bandgap is mentioned to be 1.75 eV.

5. Conclusions and outlook

The full potential of perovskite-silicon tandem solar cells can only be exploited when the devices are optimized from an optical point of view. In this manuscript, we present the results of a layer-thickness optimization for a planar inverted architecture, which has the electron-selective contacts of the top and bottom cell on the respective front sides. Because the supporting layers of the inverse architecture can be manufactured much thinner [8], the parasitic absorption in the front layers of the top cell is strongly reduced. Hence, potentially an additional cumulative ≈ 2.8 mA/cm² can be generated in the two subcells with respect to the regular architecture [14]. To achieve the maximal current densities, the front layers ITO, SnO₂ and PCBM layers must be as thin as possible. If it were possible to manufacture the supporting layers in the regular architecture just as thin as in the inverse architecture, a similar gain in current density could be achieved. When allowing larger perovskite bandgaps, we find a maximal power conversion efficiency of 30.8% at 1.73 eV perovskite bandgap.

For planar devices, reflection is by far the largest optical loss accounting for more than 7 mA/cm². Light management concepts must be applied to reduce this loss: first, antireflective measures at the front side must be taken. Secondly, light trapping concepts such as (nano)texturing must be implemented at the back side in order to enhance the absorption of c-Si for long wavelength, where it only absorbs weakly because of its indirect bandgap. Thirdly, optimized intermediate layer systems between the top and bottom cells can be investigated, similar to what was done for amorphous-nanocrystalline thin-film silicon tandem solar cells [40]. Dependent on the wavelength, these layer systems redirect the light either back into the top cell or into the bottom cell in order to facilitate current matching and hence to maximize the overall photocurrent density.

In addition, it is important to investigate how tandem cells behave at different angles of incidence and under changing spectral conditions of the incident light. If coupled to meteorological data, this would enable us to estimate the potential of a certain tandem architecture with respect to the location on Earth and the orientation of the solar cell.

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